sitegeoai

Preliminary Planning Report: Typed isolated flyover fixture

Project type: Flyover | Units: metric

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PRELIMINARY PLANNING OUTPUT ONLY. All designs, dimensions, material estimates, costs, reinforcement assumptions, foundation depths, excavation quantities, and construction methodology are conceptual. They are NOT final structural drawings, NOT legal construction approval documents, and NOT a replacement for licensed civil/structural/geotechnical engineers, surveyors, architects, local authority approvals, or code compliance checks.

2. Project Summary

Preliminary 2-lane flyover concept, 206m long

Location	Explicit synthetic location
Status	analyzed
Accuracy tier	visual
Boundary area	48,188 sqm
Alignment length	206 m
AI provider	template

3. Location Map

Interactive map available in the application. Export GeoJSON for GIS workflows.

4. 3D Model

Interactive 3D model available in the application when a design scenario has been generated.

5. Site Analysis & Terrain

Area	48,188 sqm
Perimeter	878 m
Elevation range	None - None m
Slope estimate	None%

5b. Survey Grade & Validation

Accuracy tier	visual
Database mode	sqlite
Ready for design	Yes
Ready for BOQ	Yes

Validation warnings:

• Limited GIS mode (SQLite — survey imports restricted)

• DEM is missing — cut/fill and terrain mesh are approximate

• No GCPs uploaded

Recommended next steps:

• Import survey DEM for better cut/fill estimates

• This project is GIS-level only and not suitable for final construction drawings

5c. Design Engineering Review

Validation status	warning
Validation score	90

Geometry mode	alignment
Elevation mode	flat
Planning mode	template
Alignment-based	Yes

Design validation warnings:

• Terrain elevation is flat or assumed — vertical geometry may not match site.

Recommendations:

• Run site analysis and ensure terrain sampling is available along the alignment.

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6. Input Assumptions

• All values are preliminary planning assumptions, not verified engineering design

• No verified geotechnical borehole data provided

• Traffic/usage load is preliminary

• Utility data unavailable

• Generated by mock AI provider (no API key configured)

• Deterministic template parameters used (no design instructions)

7. Proposed Design Concept

• foundation: Assumed pile/open foundation placeholder

• piers: Vertical supports

• pier_caps: Cap beams

• deck_slab: Main slab

• road_surface: Wearing coat/asphalt

• barriers: Crash barriers/parapet

8. Dimensions

length_m	206.39
deck_width_m	12.0
clearance_m	5.5
pier_spacing_m	20.0
pier_count	11
foundation_depth_m_assumed	8.0

9. Quantity Estimate / BOQ

Item	Qty	Unit	Rate	Amount
Foundation excavation	3,168.0	m3	250.0	792,000

Structural concrete M35	2,458.9	m3	8,500.0	20,900,650
Reinforcement steel	393,424	kg	75.0	29,506,800
Formwork	9,835.6	sqm	650.0	6,393,140
Wearing coat asphalt	198.1	m3	11,000.0	2,179,100
Crash barriers (both sides)	412.79	m	3,500.0	1,444,752

10. Cost Estimate

Low	60,359,412 INR
Medium	71,011,073 INR
High	92,314,394 INR
Contingency	10.0%

11. Construction Sequence & Timeline

1. survey and utility detection
2. traffic diversion planning
3. foundation construction
4. pier construction
5. girder placement
6. deck slab casting
7. road surfacing
8. safety barriers and drainage
9. testing and handover

12. Risk Checklist & Limitations

• soil bearing capacity unknown
• existing utilities unknown
• Existing underground utilities not verified
• Land ownership / acquisition status not checked
• Flood and drainage impact not assessed

13. Required Surveys and Approvals

• Local authority building/infrastructure permission
• Licensed structural engineer design and certification
• Geotechnical soil investigation report
• Topographic and utility survey
• Environmental clearance (if applicable)
• Traffic police / road authority NOC (if applicable)

14. Engineering Disclaimer

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